Even human-annotated Long-CoT-like traces fail. Inspired by Du et al. [7], we test whether human step-by-step solutions can induce long CoT. We collect human solutions for complex reasoning tasks and fine-tune LLMs on them. Figure 4 shows that human-trace training does not reproduce the long-CoT gains from distilling strong reasoning models, suggesting that human solutions aid local problem solving but may not reliably encode abstractions for long-horizon reasoning distributions.

Takeaway 1

Distillation from strong reasoning LLMs effectively imparts long CoT structures, while ICL from weak instruction models and fine-tuning on human traces yield limited gains. This underscores the importance of high-quality reasoning exemplars for robust long-chain learning in LLMs.

3 Hypothesis: Stable "Molecular Structure" in Long CoT

To understand these phenomena, as shown in Fig. 1 & 2, effective Long CoT connects nodes in logical space through reasoning behaviors, forming a stable macromolecular structure with mutually supportive components from a global perspective.

We formalize a Long CoT trace as a behavior-directed graph $G = (V, E)$, where each node $v \in V$ represents a reasoning step or edge $s = (u, v) \in E$ is annotated with a behavior type $s \to b \in \{\mathcal{D}, \mathcal{R}, \mathcal{E}\}$. For a trace corpus $\mathcal{C}$, we estimate the behavior transition $P_{\mathcal{C}}(b' \mid b)$ over consecutive edges and the marginal distribution $\pi_{\mathcal{C}}(b)$. Empirically, strong reasoning teachers produce stable $(P_{\mathcal{C}}, \pi_{\mathcal{C}})$ across models and tasks. Specifically, these include the following three major bonds[†]:

Deep Reasoning as Covalent Bonds Deep reasoning forms the bone of the thought process, analogous to covalent bonds defining a molecule's primary chain. It encodes strong logical dependencies (Step A must justify Step B), maintaining direction and continuity; breaking this bone undermines the following steps and destabilizes the answer. By contrast, "Normal Operation" corresponds to stable local bonds within each step, capturing routine computation and direct semantic expression.

Self-Reflection as Hydrogen Bonds Reflection is a key stabilizer. As proteins gain stability when chains fold and form intra-chain hydrogen bonds, reasoning stabilizes when later steps (e.g., Step 100) test, revise, or reinforce earlier premises (e.g., Step 10). These long-range links constrain drift and hallucination, turning a long sequence into a more self-consistent structure. If later checks fail to align with earlier commitments, the reasoning cannot "fold," indicating a structural logical error.

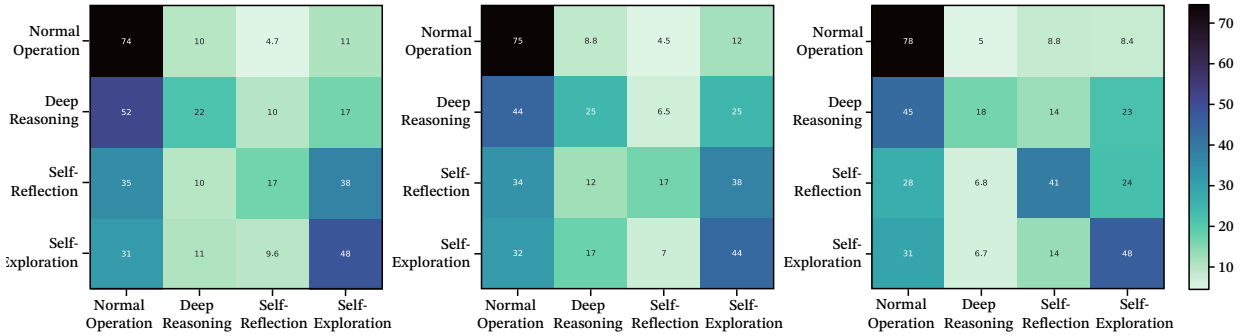
Self-Exploration as Van der Waals Forces Exploration resembles transient Van der Waals interactions: it supports abductive and inductive moves by enabling low-commitment associations in semantic space, where concepts can drift, combine, and be probed before stronger constraints are enforced.

4 Verification: Molecular Structure

4.1 Stable Bond Distribution in Long CoT

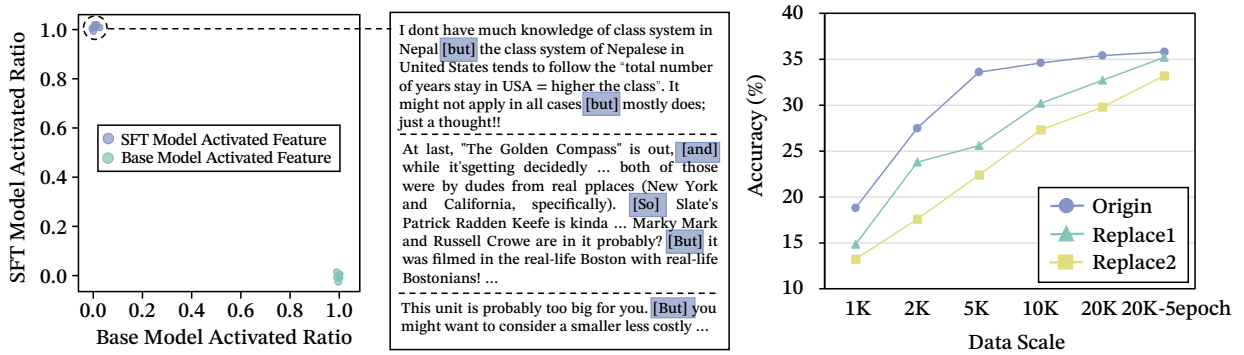
To address this, we first verify whether effective Long CoT traces show stable, macromolecular-like organization across models and tasks. As shown in Fig. 5, traces from multiple LLMs across diverse tasks yield Pearson correlations exceeding 0.9 ($p<0.001$) for over 2,000 samples. These transfer graphs stabilize with correlations above 0.95 across sampling sizes. This indicates that effective Long CoT structures rely on robust motifs: different models recover similar reasoning topologies across tasks, whereas simple human simulation or ICL cannot emulate the global bond distribution.

[†]Math definitions and proofs are in Appendix C.1.



(a) Transfer graph on DeepSeek-R1-0528. (b) Transfer graph on OpenAI-OSS-120B. (c) Transfer graph on QwQ-32B.

Figure 5 Transfer graph on three different models. Pearson correlation coefficients across models are all greater than 0.9 ($p<0.001$), when sampling examples $>$ 2,000, transfer graphs will become stable and get over 0.95 Pearson correlation between different sampling sizes.



(a) Sparse Crosscoder features with a non-shared activation ratio (>0.8) between the base and SFT models.

(b) Tokens activated by features that appear only in the SFT model and not in the base model.

(c) Performance of the SFT-trained model when keywords in QwQ's reasoning chains are replaced with synonyms at different data scales.

Figure 6 The learned features in Long CoT supervised fine-tuning.

4.2 SFT actually learns these bond structures rather than keywords.

For the SFT analysis, we consider Llama-3.1-8B-Base that is pre-trained but not instruction-tuned on Long CoT data; Then, we consider Llama-3.1-8B-Base trained on R1-distilled data as Think-SFT model obtained by supervised fine-tuning enriched with Long CoT traces.

From a representation-geometry perspective, the sparse auto-encoder analysis shows that Long CoT behavior in the SFT model is concentrated in a small set of discourse-control structures rather than being uniformly distributed across tokens. As shown in Figure 6 (a, b), we train a cross-coder sparse auto-encoder that jointly models hidden states from the base model and the Think-SFT model, and then identify features whose activation on think tokens is at least threefold higher than their average activation rate. Within this highly activated subset, many features are predominantly driven by a few connective keywords, such as *"Maybe"*, *"But / so"*, and *"Alternatively"*, indicating that the SFT process has carved out dedicated latents for managing local hypothesis revision, contrastive moves, and branch selection in long CoT traces.

Based on previous analyses, we argue that models learn the characteristic reasoning behaviors these keywords represent, not the keywords themselves. Following Chen et al. [5], we define three categories of Long CoT reasoning behaviors and test this claim using two training datasets derived from QwQ distillation data. In the first, we replace each keyword with one of four alternative variants. In the second, we remove all keywords while preserving the reasoning trajectories. We then fine-tune identical LLMs on each dataset and evaluate their Long CoT reasoning performance.

As shown in Figure 6 (c), explicit keywords like "wait" accelerate learning but are not essential. Models trained

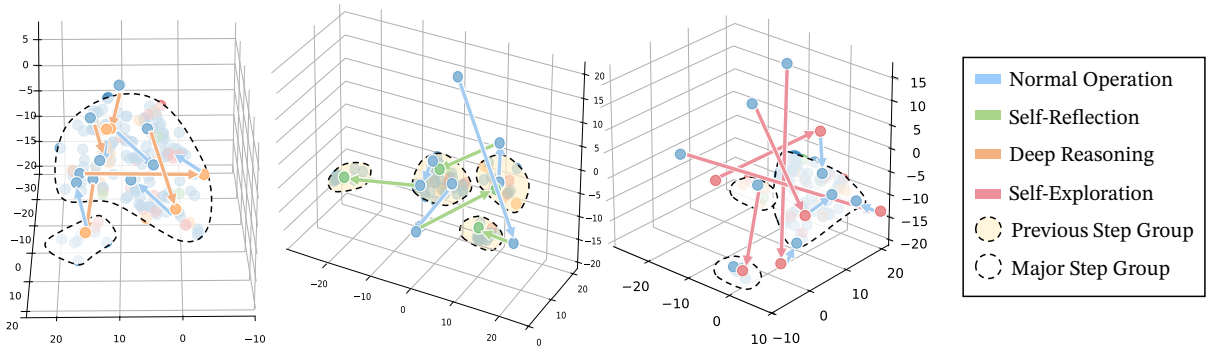
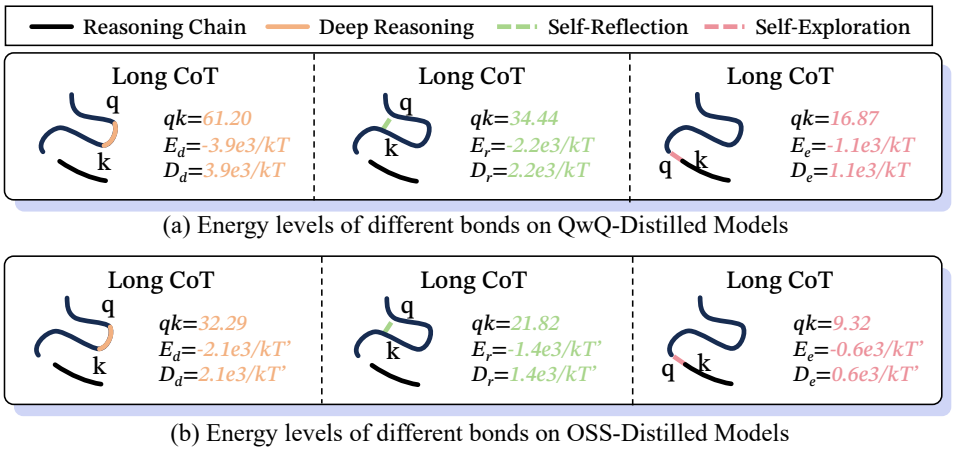


Figure 7 Verification of the "logical folding" structure on embeddings from Qwen2.5-32B-Instruct and t-SNE-based low-dimensional representations from the OpenAI-OSS-120B-generated reasoning process.



(a) Energy levels of different bonds on QwQ-Distilled Models

(b) Energy levels of different bonds on OSS-Distilled Models

Figure 8 Energy levels of different bonds across two distilled data.

without keywords, or with arbitrary alternatives, achieve comparable reasoning performance given sufficient training, provided the underlying reasoning behaviors remain intact. This reveals a fundamental insight: LLMs internalize reasoning structure of reasoning rather than surface lexical cues. Consequently, training data should prioritize the distribution of reasoning behaviors over specific keyword choices to effectively enhance model reasoning capabilities.

However, a key open question remains: do these bonds drive Long CoT structure learning, and if so, **why do explicit human imitation or random ICL distillation of these markers often fail?**

4.3 "Logical Bonding-Folding" Structure

To test the hypothesis that Long CoT is analogous to macro-molecular folding, as shown in Fig. 7, we analyzed the topology of CoT in a 3D semantic space. Each trajectory edge was classified as reflection, deep reasoning, or exploration, and its geometric properties were quantified.

Deep-Reasoning stabilizes logical cluster by Covalent Bonding. Modeled as covalent bonds, as shown in Fig. 7 (a), deep reasoning bonds mainly increase local connectivity, forming stable subdomains. After deep reasoning, 72.56% of steps remained within a group distance of less than 3 in the semantic space (generally, group-group distance >5.6).

Self-Reflection drives strong folding to previous steps by Hydrogen Bonding. Analogous to hydrogen bonds, as shown in Fig. 7 (b), self-reflection transitions fold later steps back onto earlier, semantically similar clusters rather than extending the chain linearly. Quantitatively, 81.72% of reflection steps reconnected to a previously formed cluster with high semantic similarity.